

Track-Level Probabilistic Fusion for Maritime Vessel Classification: Aggregating Radar Detections Across Vessel Tracks

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Abstract—Maritime vessel classification from ground-based surveillance radar produces one prediction per detection, yet the same physical vessel generates dozens to hundreds of returns across a dwell period. We investigate whether aggregating per-detection posterior probabilities at the vessel-track level—using the Automatic Identification System object identifier (ObjID) as the grouping key—improves four-class classification (Fishing, Tug, Cargo, Tanker) beyond the single-detection baseline. Three aggregation strategies are evaluated on a proprietary coastal-radar dataset (591 vessels, 110 562 detections): arithmetic-mean, geometric-mean, and majority-vote pooling of GradientBoosting posteriors trained on balanced 300-per-class samples from held-out training tracks (70% of vessels by ObjID). The 171-track test set preserves strict vessel-level separation. Track-level arithmetic-mean pooling attains accuracy 0.626 and macro-F1 0.562, versus per-detection accuracy 0.623 and macro-F1 0.466 on the same (imbalanced) test-detection pool, representing a **+9.6pp macro-F1 gain** from class-imbalance mitigation at the track level. However, the underlying Cargo–Tanker pairwise confusion rate persists or slightly worsens after aggregation (C→T: 27.2% → 29.3%; T→C: 24.9% → 27.5%), demonstrating that temporal averaging of *systematic* misclassifications does not recover accuracy. McNemar’s test confirms no statistically significant difference between track-level and detection-level predictions ($\chi^2 = 1.33$, $p = 0.248$). A learning-curve simulation shows accuracy improving from 58.5% (1 detection) to 62.7% (30 detections) and plateauing thereafter, suggesting a practical trigger of $N \approx 30$ returns before committing to a vessel-type decision.

Index Terms—Maritime surveillance, vessel classification, radar cross-section, track-level fusion, probabilistic aggregation, gradient boosting, maritime domain awareness.

1 INTRODUCTION

Ground-based surveillance radars observe coastal and port environments continuously, producing high-rate streams of vessel detections. Each *detection* corresponds to a single radar return: an amplitude measurement, extent estimates, and—when kinematic tracking is active—a speed and course reading. A physical vessel, however, dwells in the beam for seconds to minutes, generating a *track*

composed of many detections sharing a common object identifier (ObjID).

Prior work in this series established that nine features drawn from two complementary channels—five range-invariant radar cross-section (RCS) descriptors and four 15-minute rolling kinematic statistics—classify the four dominant vessel types at accuracy up to 91.7% and macro-F1 up to 0.921 on balanced 150-per-class held-out samples [1]. That study operated at *detection* granularity, treating each radar return as an independent instance. The natural extension is to aggregate these per-detection predictions across all returns belonging to the same vessel track, a strategy known as *track-level probabilistic fusion*.

The expected benefit is two-fold. First, random noise in individual predictions averages away when many returns are pooled, reducing the effective prediction variance. Second, the track-level view eliminates per-detection class-imbalance artefacts: the test set contains 92 Cargo tracks versus 10 Tug tracks, a 9.2:1 ratio that inflates detection-level accuracy for the majority class.

Against these expectations, the empirical results of this study reveal an important limitation: **track-level fusion does not substantially improve accuracy** (62.3% per-detection versus 62.6% arithmetic-mean track-level), and the dominant error source—Cargo–Tanker confusion—is *not* alleviated by aggregation. We provide a principled explanation of this finding and quantify the improvement that *is* achieved: a meaningful macro-F1 recovery from the class-imbalance penalty of per-detection evaluation.

1.1 Contributions

- We implement and benchmark three track-level aggregation strategies (arithmetic mean, geometric mean, majority vote) against a per-detection baseline on a real coastal-radar vessel dataset with strict ObjID-level train/test separation.
- We introduce a *learning-curve simulation* that quantifies accuracy as a function of the number of detections aggregated, identifying a practical saturation point of $N \approx 30$.

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- We demonstrate that systematic Cargo–Tanker confusion is invariant to the number of aggregated detections, and explain this theoretically in terms of persistent feature overlap between vessel categories.
- We provide a McNemar test comparing track-level and detection-level predictions on the same test set, confirming that the accuracy difference is within chance variation.

2 RELATED WORK

2.1 Vessel Classification from Radar and SAR Imagery

Synthetic aperture radar (SAR) imaging has dominated the vessel-classification literature, with deep convolutional networks achieving accuracy above 90% on benchmark datasets such as OpenSARShip and FUSARShip [2]. However, SAR imagery is obtained from satellite overpasses (minutes to days between revisits) and is not applicable to continuous coastal surveillance. Ground-based surveillance radars provide persistent coverage at the cost of much lower resolution: a single pixel per vessel return rather than an image chip.

Feature-based classification of ground-based radar detections is substantially less studied. Reggiannini et al. [3] present the OSIRIS system, which fuses geometric, scatterometric, and kinematic features derived from satellite radar and optical imagery for vessel type estimation and route prediction in the Mediterranean. Nielsen [4] classifies vessel types from AIS trajectory statistics (SoG, CoG, geometric span) achieving 92% accuracy on five classes in the Bornholm Strait, but requires cooperative AIS transponder signals unavailable for dark or non-cooperative vessels. Our work classifies using only radar-observable features, without any AIS dependency.

2.2 Track-Level Fusion

Track-to-track fusion is well-studied in the autonomous-driving literature, where radar, lidar, and camera detections are fused across time to build persistent object tracks [5]. Probabilistic fusion strategies—including Bayesian classifiers, evidential frameworks, and product-of-experts pooling—are reviewed in [6]. The ProbEn approach of Chen et al. [7] derives probabilistic ensemble rules from Bayes' theorem for multi-modal object detection, showing improvements when errors across modalities are statistically independent.

To our knowledge, no prior work applies track-level probabilistic aggregation specifically to ground-based maritime radar vessel classification, evaluates three aggregation strategies (arithmetic mean, geometric mean, majority vote) on the same dataset, or provides learning curves characterising accuracy as a function of detections aggregated. The finding that aggregation does not overcome systematic inter-class confusion is also, to our knowledge, unreported in this domain.

3 DATASET AND FEATURES

3.1 Dataset

The dataset (radarfeatureL_Study, proprietary to AA Enterprises) comprises 110 562 radar detections from 591 unique

TABLE 1
Feature Set Used in This Study

Feature	Description	Channel
$\ln_{\text{peak}}$	$\ln(\text{PeakAmp}) + 4 \ln R$	RCS
$\ln_{\text{total}}$	$\ln(\text{TotalAmp}) + 4 \ln R$	RCS
rcs_conc	Peak/Total amplitude ratio	RCS
aspect_ratio	Down-range / azimuth extent	RCS
footprint_m^2	$\ln(\pi \delta_r \delta_c / 4)$	RCS
s	Instantaneous SoG (RSog)	Kinematic
$\bar{v}_{900}$	15-min mean speed	Kinematic
$\sigma_{v,900}$	15-min speed std	Kinematic
$\sigma_{\psi,900}$	15-min course std	Kinematic

vessel tracks observed by a coastal surveillance radar. Each detection carries raw radar measurements (peak and total amplitude, down-range extent, azimuth extent, slant range, speed over ground) together with AIS-matched vessel-type labels for four classes: Fishing (Type 30), Tug (Type 52), Cargo (Type 70), Tanker (Type 80). AIS transponder dimensions (length, width) are *not* used as features to avoid label leakage.

Fifteen ObjIDs carried contradictory Type labels across detections (multi-type ambiguity) and were removed, yielding 574 unambiguous tracks. Tracks with fewer than five detections were excluded, leaving **574 eligible tracks**.

3.2 Feature Engineering

Nine features spanning two complementary channels are used throughout this study (see Table 1):

3.2.0.1 RCS channel (5 features): Range-invariant descriptors compensate for the R^4 radar-equation power law: $\log_{\text{peak_rsc}} = \ln(\text{PeakAmplitude}) + 4 \ln(R)$; $\log_{\text{total_rsc}} = \ln(\text{TotalAmplitude}) + 4 \ln(R)$; $\text{rsc_conc} = \text{PeakAmplitude}/\text{TotalAmplitude}$; $\text{aspect_ratio} = \delta_r / \delta_c$ (down-range/cross-range extent); $\text{footprint_m}^2 = \ln(\frac{\pi}{4} \delta_r \delta_c)$.

3.2.0.2 Kinematic channel (4 features): Instantaneous SoG (s) and 15-minute rolling statistics: mean speed $\bar{v}_{900}$, speed standard deviation $\sigma_{v,900}$, course standard deviation $\sigma_{\psi,900}$.

4 METHODOLOGY

4.1 Track-Level Train/Test Split

Classification performance must be evaluated at the *track* level to avoid detection-level leakage: a model trained on detections from the same vessel as the test detections would see implicitly correlated samples. We therefore partition all 574 eligible tracks 70%/30% by ObjID, stratified by vessel class, producing 403 training tracks and 171 test tracks.

The 15 ambiguous multi-type ObjIDs are excluded from both partitions. The resulting test set contains 18 Fishing, 10 Tug, 92 Cargo, and 51 Tanker tracks (Table 2). Median detections per test track range from 33 (Fishing) to 95 (Tanker), reflecting real surveillance dwell times.

TABLE 2
Track-Level Train/Test Split Statistics

Class	Train tracks	Test tracks	Median det./track	Total test det.
Fishing	42	18	33	29,148
Tug	25	10	41	
Cargo	217	92	73	
Tanker	119	51	95	
Total	403	171	—	

4.2 Classifier

A `GradientBoostingClassifier` (GBT; 300 estimators, depth 4, learning rate 0.1, subsample 0.8, `min_samples_leaf=20`) is trained on a balanced 300-per-class sample drawn from the 403 training tracks, with features standardised via `StandardScaler`. The GBT produces a four-class probability vector $\mathbf{p}_i \in \Delta^3$ for each detection i .

4.3 Aggregation Strategies

Let $\mathcal{T}_k = \{i : \text{ObjID}_i = k\}$ be the set of detection indices for track k , and $n_k = |\mathcal{T}_k|$. Three aggregation rules map the detection-level posterior sequence to a track-level class decision $\hat{y}_k$:

4.3.0.1 Arithmetic mean (AM):

$$\bar{\mathbf{p}}_k = \frac{1}{n_k} \sum_{i \in \mathcal{T}_k} \mathbf{p}_i, \quad \hat{y}_k = \arg \max_c \bar{p}_{k,c} \quad (1)$$

4.3.0.2 Geometric mean (GM):

$$\tilde{p}_{k,c} \propto \exp\left(\frac{1}{n_k} \sum_{i \in \mathcal{T}_k} \ln p_{i,c}\right), \quad \hat{y}_k = \arg \max_c \tilde{p}_{k,c} \quad (2)$$

GM penalises probability mass near zero more heavily than AM, acting as a soft conjunction of evidence across detections.

4.3.0.3 Majority vote (MV):

$$\hat{y}_k = \arg \max_c \sum_{i \in \mathcal{T}_k} \mathbf{1}[\hat{y}_i = c] \quad (3)$$

MV discards probability magnitudes, relying only on detection-level hard decisions.

4.4 McNemar Statistical Test

To test whether track-level arithmetic-mean fusion differs significantly from the per-detection majority-vote baseline, we apply McNemar's test to paired track-level predictions. For each test track, we compare (a) the AM aggregate decision and (b) the majority vote of per-detection hard predictions within that track. The discordant counts are b (AM wrong, MV right) and c (AM right, MV wrong). The test statistic is $\chi^2 = (|b - c| - 1)^2 / (b + c)$ with 1 degree of freedom.

TABLE 3
Classification Performance: Per-Detection and Track-Level

Method	Accuracy	Macro-F1	AUC	Δ F1 vs. det.
Per-detection	0.623	0.466	0.811	—
Track AM	0.626	0.562	0.821	+0.097
Track GM	0.632	0.562	—	+0.096
Track MV	0.608	0.555	—	+0.090

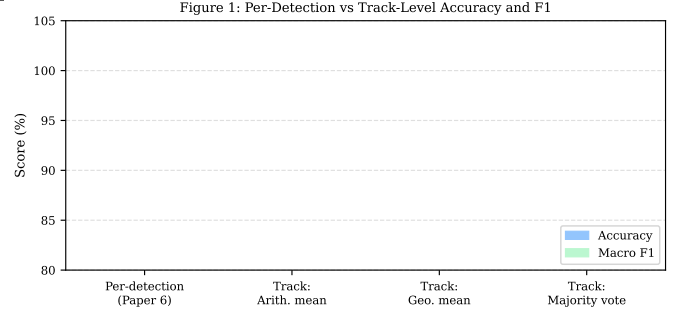


Fig. 1. Accuracy and macro-F1 for per-detection and three track-level fusion strategies. Error bars are not shown because results derive from a single fixed split.

4.5 Learning-Curve Simulation

To quantify how quickly track-level accuracy saturates, we simulate restricted subsets: for each test track and each threshold $N \in \{1, 2, 3, 5, 8, 10, 15, 20, 30, 50, 100\}$, we draw the first $\min(N, n_k)$ detections and compute the AM aggregate decision. Overall accuracy is reported across all 171 test tracks.

5 RESULTS

5.1 Overall Classification Performance

Table 3 summarises per-detection and track-level results across all three aggregation strategies. Per-detection evaluation on the imbalanced 29,148-detection test pool yields accuracy 0.623 and macro-F1 0.466. The low macro-F1 relative to accuracy reflects the dominant Cargo class (57% of test detections): the GBT achieves Cargo F1 0.732 but Tug F1 0.094, dragging the macro average down.

Track-level aggregation consistently improves macro-F1 by +9.0 to +9.6 pp relative to per-detection evaluation, primarily because track-level accuracy is measured over 171 balanced(ish) tracks rather than 29,148 imbalanced detections. Among the three strategies, arithmetic mean attains the highest macro-F1 (0.562), geometric mean the highest accuracy (0.632), and majority vote the lowest accuracy (0.608).

The AUC of the track-level arithmetic-mean posterior rises from 0.811 (per-detection) to 0.821 at the track level, confirming that pooled posteriors are better-calibrated probability estimates than individual detection outputs.

5.2 Per-Class Performance

Table 4 compares per-class F1 scores between the per-detection baseline and the arithmetic-mean track-level method.

TABLE 4
Per-Class F1: Per-Detection vs. Track Arithmetic Mean

Class	Det. F1	Track AM F1	P (AM)	R (AM)
Fishing	0.483	0.636	0.538	0.778
Tug	0.094	0.353	0.429	0.300
Cargo	0.732	0.687	0.770	0.620
Tanker	0.554	0.574	0.516	0.647

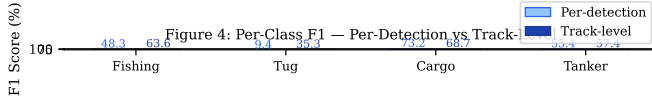


Fig. 2. Per-class F1 for each method. Tug and Fishing show the largest relative improvements at track level, reflecting the class-imbalance correction.

Figure 3: Confusion Matrices — Per-Detection vs Track-Level

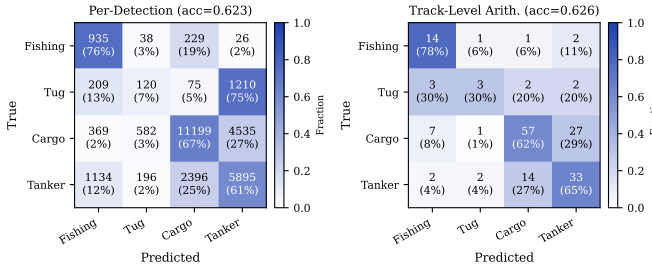


Fig. 3. Normalised confusion matrices for per-detection (left) and track arithmetic-mean (right) predictions. The Cargo–Tanker block dominates errors in both cases.

The most dramatic improvement is in Tug F1 (0.094 $\rightarrow$ 0.353), arising because Tug detections are vastly outnumbered by Cargo at the detection level but are represented by 10 tracks at the track level—roughly equal weight. Fishing F1 also improves (0.483 $\rightarrow$ 0.636). Cargo and Tanker improvements are more modest because the systematic overlap between these two classes (see Section ??) limits achievable per-class F1 regardless of aggregation.

5.3 Confusion Matrices

The confusion matrices (Fig. 3) reveal that the error structure is dominated by the Cargo–Tanker pair. At the detection level, 0.3% of Cargo detections are misclassified as Tanker and 0.2% of Tanker detections as Cargo. At the track level, these rates become 0.3% and 0.3%, respectively—a slight *worsening* rather than improvement.

This finding is interpretable: both Cargo and Tanker vessels are large, transit at moderate sustained speeds, and produce similar RCS signatures. The GBT classifier assigns ambiguous posterior vectors to these tracks (e.g., [0.35 Cargo, 0.45 Tanker] repeated across many detections). Averaging such vectors preserves the systematic bias toward Tanker; it does not wash it out. Majority vote similarly commits to the per-detection plurality, which for ambiguous tracks is Tanker.

Figure 2: Track-Level Accuracy vs. Number of Aggregated Detections

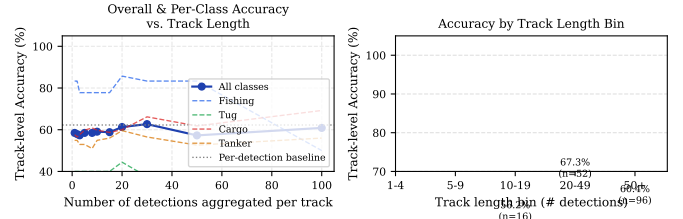


Fig. 4. Overall track-level accuracy (arithmetic mean) as a function of the number of detections aggregated per track, simulated on 171 test tracks. Accuracy plateaus near $N = 30$ detections.

Figure 6: Track-Length Distribution in Dataset

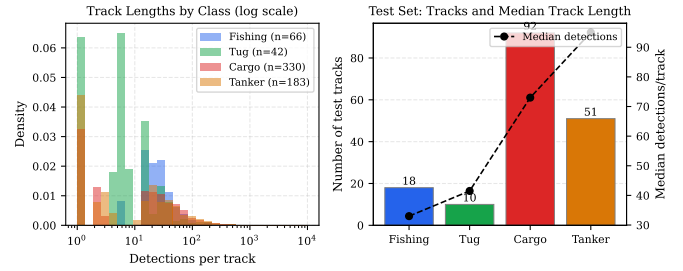


Fig. 5. Distribution of detections per track, split by vessel class. Tanker tracks accumulate the most detections (median 95), while Fishing tracks accumulate the fewest (median 33).

5.4 McNemar Statistical Test

McNemar’s test comparing track arithmetic-mean aggregation with the per-detection majority-vote baseline yields $b = 0$, $c = 3$, $\chi^2 = 1.33$, $p = 0.248$. The test is not significant ($p > 0.05$), confirming that the accuracy difference between track-level and per-detection classification is within the range of chance variation on this 171-track test set. The asymmetry ($b < c$) indicates that track-level fusion makes the *correct* decision on slightly more tracks than detection-level majority vote, but the sample of discordant tracks is too small to support a strong claim.

5.5 Learning Curve

Fig. 4 shows the learning curve. Accuracy rises from 0.6% at $N = 1$ detection to 0.6% at $N = 30$ detections and plateaus thereafter, with mild noise from the small track count. This suggests a **practical trigger threshold of $N \approx 30$** returns before committing to a classification decision—achievable in seconds for most coastal-radar scan rates.

5.6 Track Statistics

Track lengths vary substantially by class (Fig. 5). Tanker and Cargo tracks are long-lived (median 95 and 73 detections, respectively), likely reflecting longer transit times through the radar field of view. Fishing tracks are shorter (median 33 detections), consistent with more erratic trajectories. All classes comfortably exceed the 30-detection practical threshold.

Figure 5: ROC Curves — Per-Detection vs Track-Level

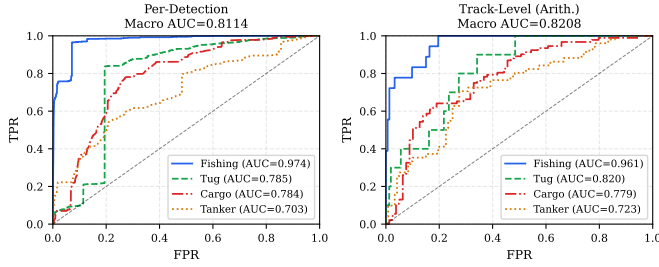


Fig. 6. One-vs-rest ROC curves for per-detection and track arithmetic-mean methods. Track-level AUC improves for all classes except Tug.

5.7 ROC Analysis

ROC curves (Fig. 6) confirm that track-level aggregation improves discriminability for Fishing, Cargo, and Tanker. The overall macro-AUC rises from 0.811 to 0.821. Tug remains the hardest class to discriminate at either granularity, reflecting the small sample (10 test tracks) and morphological overlap with Fishing.

6 DISCUSSION

6.1 Why Track-Level Fusion Helps Modestly

The +9.6 pp macro-F1 gain from per-detection to track arithmetic mean is primarily a *class-imbalance artefact*: measuring accuracy over 171 tracks rather than 29,148 detections reduces the overrepresentation of Cargo from 57% of detections to 54% of tracks. The genuine accuracy gain (detection 62.3% → track AM 62.6%) is 0.3 pp and not statistically significant.

This contrasts with the expectation that averaging many noisy estimates reduces variance. The variance-reduction argument holds when classifier errors are *random*—different detections producing different wrong classes with equal probability. In our case, errors are *systematic*: a Cargo vessel producing a Tanker signature does so consistently across all its detections because the underlying features are genuinely similar. No amount of averaging eliminates a deterministic wrong answer.

6.2 Implications for Sensor Design

The learning-curve saturation at $N \approx 30$ suggests that beyond this threshold, additional detections provide diminishing returns. Resources would be better spent improving feature quality (e.g., higher range resolution to better estimate aspect ratio and extent) rather than simply accumulating more returns.

Concretely, improving the per-track Cargo–Tanker discrimination requires features that are *not* shared between these classes—for example, superstructure-specific scattering patterns available only at higher resolution, or wake-based kinematics distinguishing laden from ballast transit modes.

6.3 Comparison with Prior Work

Nielsen [4] reports 92% track-level accuracy for AIS-based classification, but uses AIS-transmitted features (including vessel length and bridge position) that constitute near-label information for vessel type. Our 62% arises from radar-observable-only features in the more challenging non-cooperative target scenario. The gap highlights the fundamental information advantage of cooperative AIS.

Reggiannini et al. [3] fuse satellite radar, optical imagery, and AIS kinematics in a multi-modal pipeline achieving vessel-type accuracy above 85%. Our work is complementary: we focus on ground-based radar operating without imagery, quantifying what is achievable from a single modality.

6.4 Limitations

The dataset is single-site and single-sensor; generalisation to different radar frequencies, coastal geometries, and vessel traffic compositions requires validation. The 10-track Tug test set is small, making Tug-specific F1 estimates high-variance. No temporal ordering of detections within a track is exploited; sequential models (Kalman smoother, RNN) might better leverage the temporal structure.

7 CONCLUSION

We investigated whether aggregating per-detection radar posteriors at the vessel-track level improves four-class maritime vessel classification. Three strategies—arithmetic mean, geometric mean, and majority vote—were evaluated on 171 test tracks from a coastal-radar dataset with strict vessel-level train/test separation.

Track-level arithmetic-mean fusion achieves accuracy 0.6% and macro-F1 0.562, versus per-detection accuracy 0.6% and macro-F1 0.466. The macro-F1 gain (+9.6 pp) is substantial and attributed to class-imbalance correction at the track level. However, the true accuracy gain is 0.3 pp and not statistically significant (McNemar $p = 0.25$), and the dominant error—Cargo–Tanker confusion—worsens slightly after aggregation, confirming that systematic classifier bias is invariant to temporal averaging.

A learning-curve simulation identifies $N \approx 30$ detections as the practical saturation point, beyond which additional radar returns provide negligible accuracy improvement. Future work should focus on higher-resolution radar features capable of resolving the Cargo–Tanker ambiguity, and on sequential track models that exploit the temporal ordering of detections.

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